

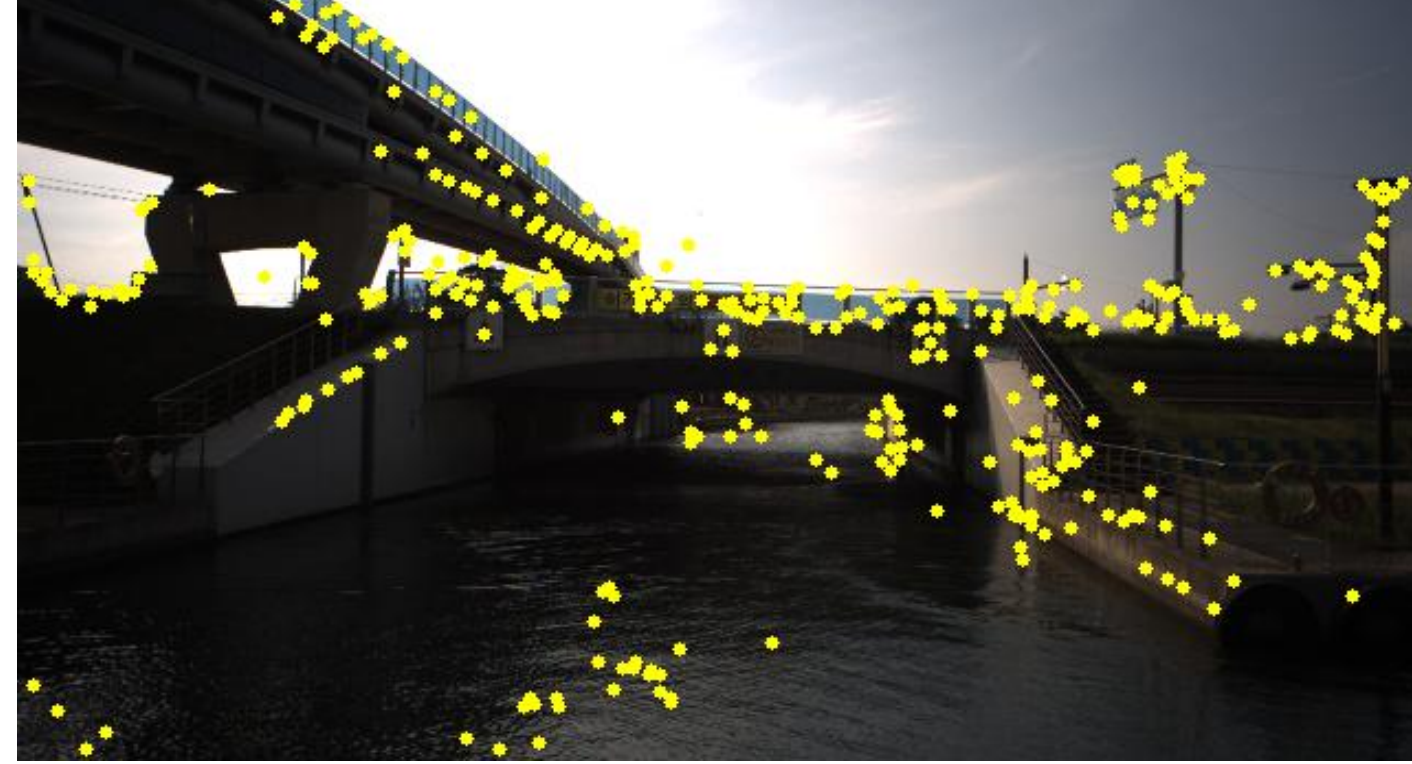
Good Deep Features To Track: Self-Supervised Feature Extraction and Tracking for Visual Odometry

Motivation and Objective

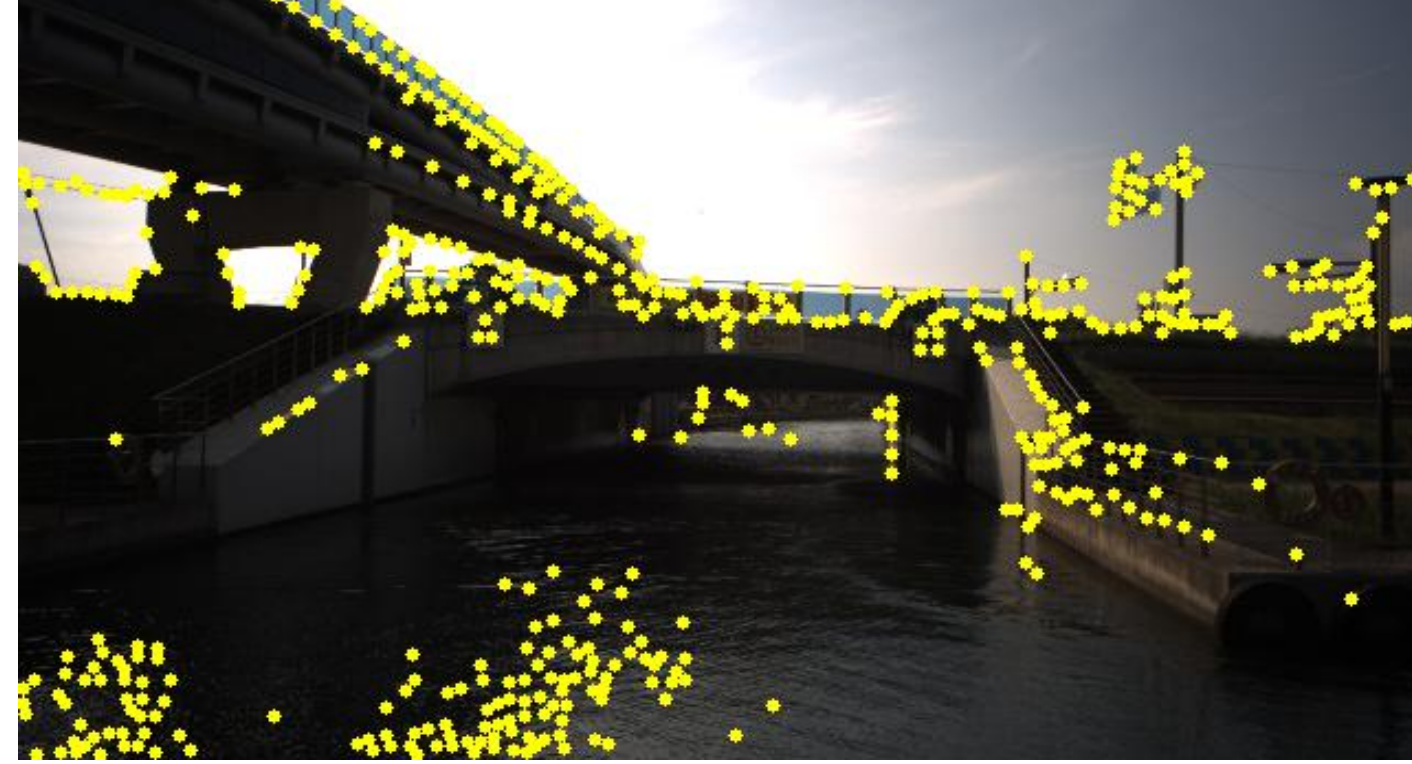
Visual-based robot localization has achieved remarkable success with a wide range of approaches tailored to specific applications. However, the performance of state-of-the-art approaches often degrades in large-scale and outdoor environments during a long-term operation, where a variety of challenges arise, such as dynamic lighting conditions and objects, low-parallax features and textureless surfaces. These factors dramatically impact the quality and robustness of one of the most crucial steps in visual-based methods: feature extraction and tracking. Figure 1 exemplifies this issue in a maritime environment, where illumination and scene appearance can vary frequently even at the same location.



(a) Original Image.



(b) Extracted SIFT [1] features.



(c) Extracted GFTT [2] features.



(d) Extracted (original) SuperPoint [3] features.

Fig.1 Extracted features (yellow) of visual odometry with different state-of-the-art approaches using the Pohang dataset (Pohang 04) [4].

Generally, a well-distributed coverage of tracked features across the entire image is expected to provide better geometric constraints. Assuming that the scale information is given, distant features help constrain rotation, while nearby features are essential for accurately resolving translational motion due to their stronger parallax [5]. Keep this in mind, classic feature extractors such as SIFT and GFTT tend to only focus on texture-rich and well-lit regions, resulting in densely clustered features and poor coverage across the image (see Fig.1b and 1c). In contrast, deep learning-based methods such as SuperPoint [3] produce more uniformly distributed results, even in low-texture or challenging lighting conditions (see Fig.1d). However, since pre-trained model often degrades during inference when exposed to out-of-distribution data, obtaining high-quality features that can be robustly tracked to ensure accurate motion estimation in diverse real-world environments remains an open research question.

Although many methods exist to improve feature quantity — such as sophisticated feature matching and filtering [5], optimizing camera attributes (e.g., exposure time) [6], and removing dynamic objects [7] using semantic information [8] — they still struggle to generalize reliably for long-term motion estimation in diverse outdoor environments. Moreover, we observe that the distinction between “good” and “poor” visual features is not always clear-cut, as it depends not only on geometric properties but also on factors such as scale and temporal consistency. In visually degraded scenarios, features previously considered “poor” may still be trackable over short time windows and, in some cases, are sufficient to prevent visual odometry (VO) failure.

Given these observations, we aim to improve the performance and generalization of the deep learning-based feature extractor and tracker by leveraging self-supervised learning with well-designed loss terms that provide situational feedback on what constitutes “**good deep features to track**”. We chose SuperPoint [3] and SuperGlue [9] as our backbone networks compared to other discriminative models, as they are capable of capturing structurally and semantically meaningful features thanks to their adaptive (e.g., homographic adaptation) training strategy.

Unlike [10], which treats the VO as a black box and only tunes hyperparameters (e.g., grid size) from a system-level perspective, we argue that it is (more) important to focus on the core components of a VO. Our hypothesis is supported by preliminary results, which show that the proposed method can effectively update the model parameters and force the models to ignore unstable or uninformative features.

Method and System Overview

Fig.2 illustrates the proposed self-supervised learning pipeline in our work. We employed RTAB-Map [11] as the baseline visual odometry, where the feature extractor and tracker are replaced by SuperPoint [3] and SuperGlue [9].

During training, estimated relative pose, stereo and temporal correspondences are used to evaluate reprojection errors, which serve as geometric consistency cues. Features that are well-tracked and geometrically reliable are identified as “good” and used to compute a total weighted loss:

$$L = \underbrace{w_i * L_i(X, Y)}_{\text{cross-entropy loss for image } I} + \underbrace{w'_i * L_i(X', Y')}_{\text{cross-entropy loss for warped image } I'} + \underbrace{w_{pk} * L_{pk}}_{\text{peaky loss}} + \underbrace{w_d * L_d(\mathcal{D}, \mathcal{D}', \mathcal{S})}_{\text{hinge loss for descriptor map } \mathcal{D}}$$

where the loss terms $L_i(X, Y)$ and $L_i(X', Y')$ penalize non-keypoints score map X with their labels Y from image $[X, Y] \in I$, and the corresponding warped image $[X', Y'] \in I'$. We use a peaky loss L_{pk} to penalize inconsistent or diffuse feature responses in the score maps, which are often caused by blur or ambiguous textures. In addition, a hinge loss L_d is employed to enforce descriptor discriminability by maximizing the distance between non-matching features and minimizing it for matching pairs.

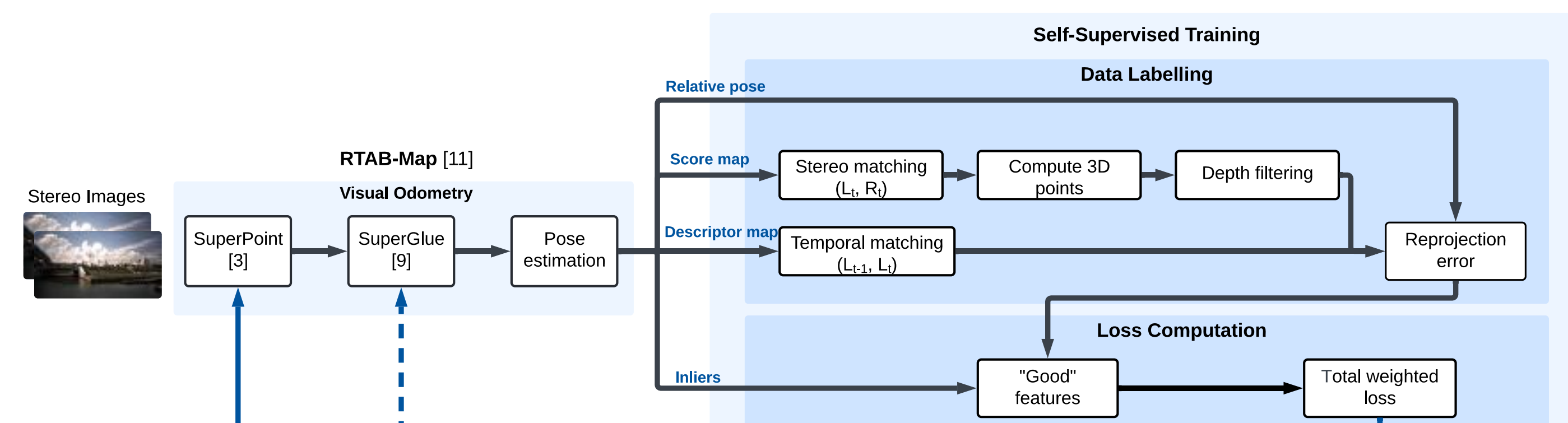


Fig.2 Proposed self-supervised pipeline to train the SuperPoint [3] and SuperGlue [9] models within a well-established visual SLAM framework RTAB-Map [11]. In the current work, only the SuperPoint model is trained, while SuperGlue remains fixed.

Preliminary Result: Feature Extraction

Fig.3 and Fig.4 present qualitative results of feature extraction using the pre-trained SuperPoint model and the proposed self-supervised version.

Compared to the former, features extracted by the self-supervised model tend to be more equitably spread across expected regions in the image, such as rigid structures, while suppressing features from less reliable areas like terrain and water surfaces.

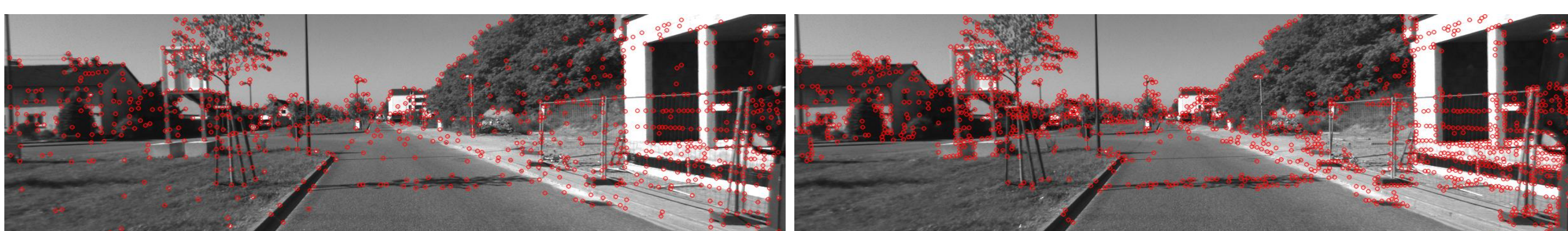


Fig.3 Extracted visual features (red) of KITTI data (seq. 06) [12] using original pre-trained (left) and self-supervised (right) SuperPoint model.



Fig.4 Extracted visual features (purple and green) of Pohang data [4] in different scenes and datetime using original pre-trained (top row) and self-supervised (bottom row) SuperPoint model.

Preliminary Result: Trajectory Plots

To validate the proposed method, we use both the pre-trained and self-supervised SuperPoint models as feature extractors within the RTAB-Map framework to estimate the vessel's trajectory. In addition, we employ the trajectory estimation using ORB [13] features as a baseline method for comparison. We evaluate all methods on the Pohang dataset [4], which is divided into sequences recorded in narrow and wide water canals, during both daytime and nighttime (see Fig. 4). The results are presented in Fig.5.

In narrow areas (Fig.4a and 4c) where most features can be acquired from infrastructure objects during daytime, the proposed model outperforms the others. A similar trend is observed during nighttime, although all VO methods experience degradation due to insufficient visual features.

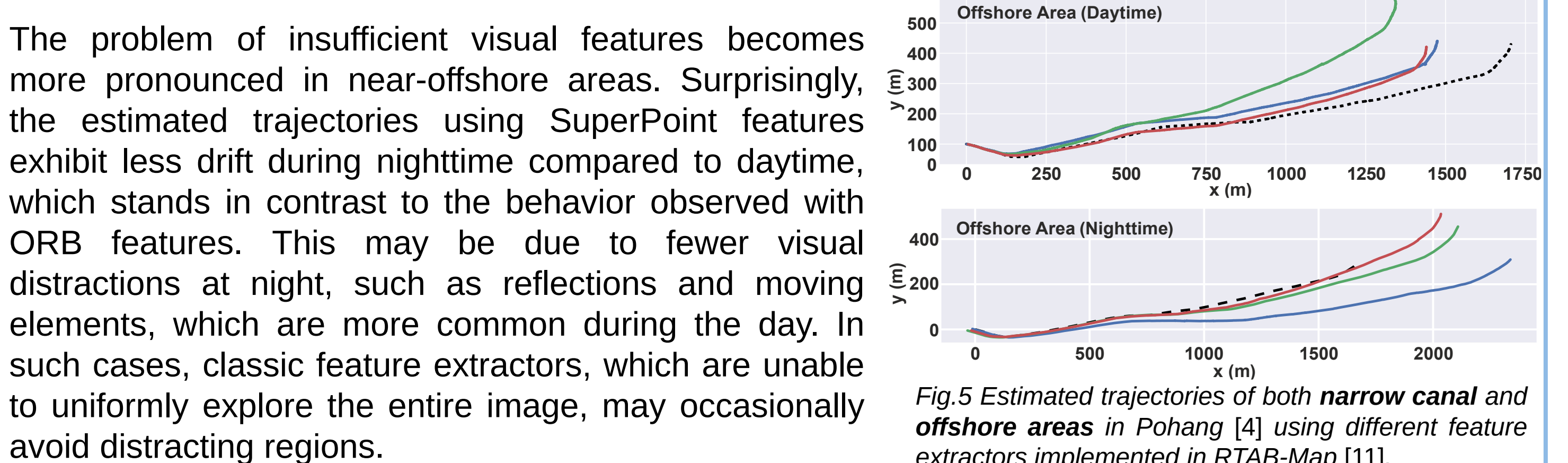


Fig.5 Estimated trajectories of both narrow canal and offshore areas in Pohang [4] using different feature extractors implemented in RTAB-Map [11].

Conclusion, Limitation, and Open Questions

This work aims to enhance the generalization of deep learning-based feature extraction and tracking within a self-supervised training framework. In the early stage of this research, we successfully demonstrate a proof of concept, showing that the model can be effectively improved using simple loss terms—without relying on label-intensive training procedures.

However, we did not train the feature matcher (e.g., SuperGlue), which remains pre-trained on the original SuperPoint. Our current results are also not yet sufficient to fully answer the question of what defines “**good deep features**” to track.

Additionally, since vision-based localization faces more challenges in outdoor environments than modalities like GNSS or LiDAR, we aim to explore how VO can be better designed and integrated into a multisensory state estimator (see Fig.6) for resilient, long-term robot navigation.

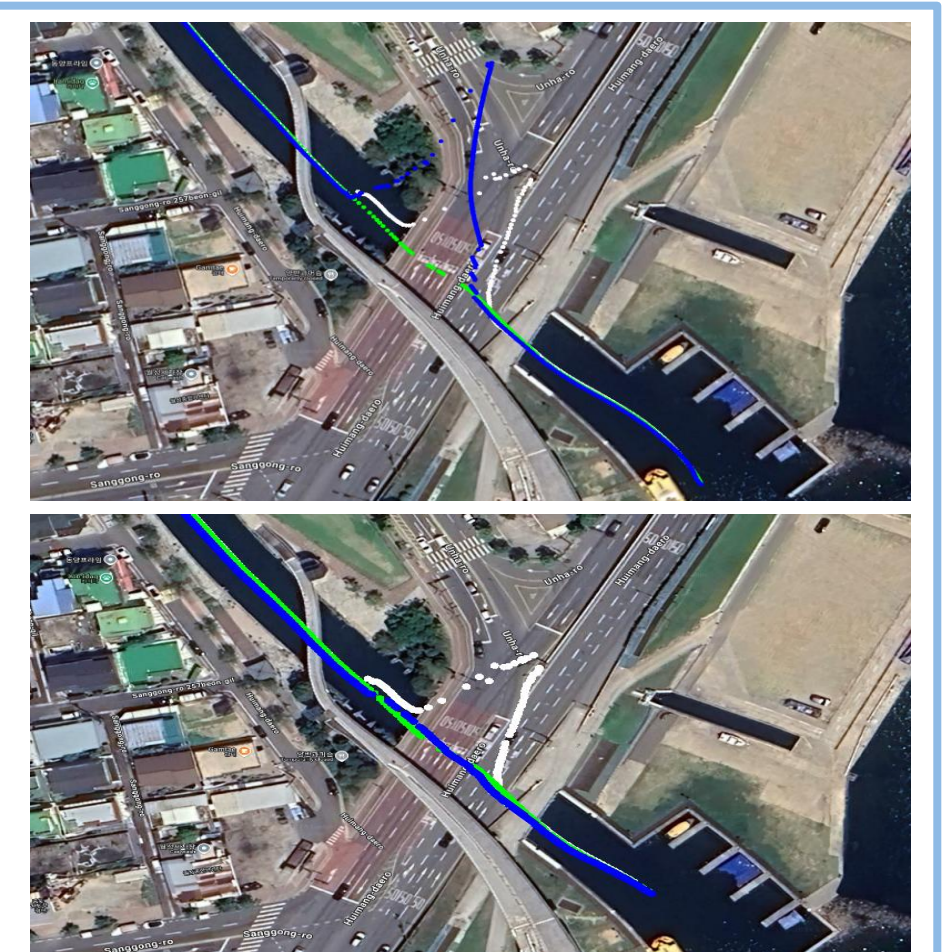


Fig.6 Demonstration of robot localization using gnssFGO [14] without (top) and with (bottom) visual odometry. The green, white, and blue trajectories represent the baseline, GNSS standalone, and gnssFGO estimates.

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